

ARTIFICIAL INTELLIGENCE IN FINTECH / WEEK 06

From Gross Returns to Net Wealth

Keep the Week 4 strategy fixed, then identify when trades occur and how stated fees change its results. Market prices are real observations; all cost rates below are teaching assumptions, not measured brokerage charges or fills. No weighting method is selected.

Issue	Question	Page
1	What can daily prices tell us about trading costs?	3
2	How much does one basis point cost?	4
3	How do different commissions charge the same order?	5
4	Why is the half-spread used for one trade?	6
5	When does adding slippage count the spread twice?	7
6	Which date carries the change in position?	8
7	Which entry and exit trades must be charged?	9
8	How does a dated trade change net return?	10
9	Why can the wealth gap differ from paid fees?	11
10	Can the entire cash balance buy stock before paying a fee?	12
11	What changes when execution is one Close later?	13
12	Does trading less often necessarily save more money?	14
13	How sensitive is wealth to the assumed cost rate?	15
14	What cost rate removes the historical gain?	16
15	How should cash and buy-and-hold be charged?	17
16	What must be checked before interpreting a net backtest?	18

Use the examples in order. Each issue ends with two discussion questions, not a separate submission. Market observations and teaching assumptions are identified separately.

SHARED DATA AND SETTINGS

Prepare the common example

AAPL, January 2-June 28, 2024; provider Close, USD/share. Signal: positive five-interval price return, requiring six closes. Decision after Close t , assumed execution at Close $t+1$, first earned return ending $t+2$. Start with USD 1,000 in cash, allow fractional shares, exclude dividends, taxes and cash interest. No forced terminal sale or new final-Close trade.

Before running: Which first and last Close define the common price interval?

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import yfinance as yf
aapl = yf.Ticker("AAPL").history(start="2024-01-02",
                                  end="2024-07-01", auto_adjust=False, actions=True)
aapl.index = aapl.index.strftime("%Y-%m-%d")
close = aapl["Close"]
assert close.index.is_unique and close.index.is_monotonic_increasing
assert np.isfinite(close).all() and (close > 0).all()
signal = (close / close.shift(5) - 1).gt(0).astype(float)
pd.DataFrame({"Date": close.index[[0,-1]],
              "Close (USD/share)": close.iloc[[0,-1]].round(2).values})
```

Date	Close (USD/share)
2024-01-02	185.64
2024-06-28	210.62

The first five signals are set to cash because their five-interval features are unavailable. Later calculations use all 124 prices and 123 price-return intervals. Daily data do not establish that a submitted order would fill at Close.

The main cost model retains the course's linear approximation: cost rate $\times$ exposure change $\times$ opening wealth. It is not exact self-financing trading. Issue 10 explains the difference. Costs are recorded on the assumed execution date, not delayed to the next return-ending date.

ISSUE 01

What can daily prices tell us about trading costs?

A daily Close is a market-data observation. It is not a record of our order, our execution price or our broker's fee. Keep that distinction when calling a result net return.

Before running: Will this response contain our order size, commission or fill price?

```
shown = aapl.loc[["2024-01-10", "2024-01-11"], ["Open", "Close"]]  
shown = shown.rename(columns={"Open": "Open (USD)", "Close": "Close (USD)"})  
shown.insert(0, "Date", shown.index)  
shown.round(2)
```

Date	Open (USD)	Close (USD)
2024-01-10	184.35	186.19
2024-01-11	186.54	185.59

January 11 Open is USD 186.54/share and Close is USD 185.59/share. They describe different times. Neither proves the price received by a particular trader.

A fee schedule can support a quoted commission. A timestamped bid and ask can support a quoted spread. Order and execution records are needed to connect a reference price to an actual fill. This lesson has only daily observations and explicit cost assumptions.

Discuss

1. A student calls the January 11 Open-to-Close change slippage. Which order, reference time and execution record are missing?
2. If a broker advertises zero commission, what other execution costs could still matter? Identify the records needed to measure them.

Sources: [1], [2], [3], [4]. See the references page.

ISSUE 02

How much does one basis point cost?

A basis point (bp) is 0.01 percentage point. Convert the rate once, then multiply by the value traded—not automatically by the entire account.

Let b be a cost rate in bp, c its decimal form, Q the traded value in USD and F the fee in USD. Buying one AAPL share at the January 11 Close gives Q equal to that Close. The rate is an assumption.

$$c = \frac{b}{10000}, \quad F = cQ$$

$$c = \frac{4}{10000} = 0.0004, \quad F \approx 0.0004 \times 185.59000 \approx 0.07424 \text{ USD}$$

Before running: Does 100 bp mean 1% or 100%? Is the 4 bp fee more or less than one US cent?

```
price = close.loc["2024-01-11"]
bp = np.array([1, 4, 10, 100])
pd.DataFrame({"Assumed rate (bp)": bp,
              "Rate (%)": bp / 100,
              "One-share fee (USD)": np.round(price * bp / 10000, 5)})
```

Assumed rate (bp)	Rate (%)	One-share fee (USD)
1	0.01	0.01856
4	0.04	0.07424
10	0.1	0.18559
100	1.0	1.8559

At 4 bp, this one-share fee is USD 0.07424, about 7.42 cents. A 4% fee would be 100 times larger. The fee is hypothetical; the Close is observed.

Discuss

1. For ten shares at the same reference price, what changes in Q and F under this proportional rate?
2. If only one quarter of the account is traded, why is subtracting 4 bp from the entire account return incorrect?

Sources: [2], [5]. See the references page.

ISSUE 03

How do different commissions charge the same order?

A fixed fee depends on charged orders, a per-share fee on share count, and a proportional fee on traded value. They cannot be substituted using one universal rate.

Use the same January 11 Close and one order. Compare assumed fees of USD 1/order, USD 0.005/share and 4 bp of traded value. Let n be shares, C the reference Close and $Q = nC$. These are separate fee alternatives, not three charges to add.

$$F_{\text{fixed}} = 1, \quad F_{\text{share}} = 0.005n, \quad F_{\text{value}} = 0.0004nC$$

$$F_{\text{value}} \approx 0.0004 \times 10 \times 185.59000 \approx 0.74236 \text{ USD}$$

Before running: Which fee stays unchanged when the single order grows from one share to 100 shares?

```
shares = np.array([1, 10, 100])
pd.DataFrame({"Shares": shares, "Fixed (USD)": 1.0,
             "Per-share (USD)": shares * 0.005,
             "Value-based (USD)": np.round(shares * price * 0.0004, 4)})
```

Shares	Fixed (USD)	Per-share (USD)	Value-based (USD)
1	1.0	0.005	0.0742
10	1.0	0.05	0.7424
100	1.0	0.5	7.4236

For 10 shares, the alternatives charge USD 1.00, USD 0.05 and USD 0.7424. Splitting an order can increase a per-order fee without changing the total number of shares. Real schedules may also impose minimums or caps, absent here.

Discuss

1. What happens to each fee if ten shares are bought as ten separately charged one-share orders at the same price?
2. Why would expressing a USD 1 fee as a constant bp rate misstate costs for different order sizes?

Sources: [2], [5]. See the references page.

ISSUE 04

Why is the half-spread used for one trade?

An immediate buyer normally faces the ask; an immediate seller faces the bid. Their difference is the full quoted spread. Relative to their midpoint, each side is half that spread.

This half-spread interpretation requires a contemporaneous quote and a midpoint reference. Close is not that midpoint. We therefore use an assumed 4 bp full spread only as a cost allowance, not as reconstructed historical quotes.

$$b_{\text{half}} = \frac{b_{\text{full}}}{2}, \quad \frac{4}{2} = 2 \text{ bp}$$

Before running: At a 2 bp one-way allowance, does the cost per share change when the reference Close changes?

```
reference = close.loc[["2024-01-11", "2024-01-17"]]
pd.DataFrame({"Date": reference.index,
              "Close (USD/share)": reference.round(2),
              "Allowance (USD/share)": (reference * 2 / 10000).round(5)})
```

Date	Close (USD/share)	Allowance (USD/share)
2024-01-11	185.59	0.03712
2024-01-17	182.68	0.03654

The January 11 allowance is USD 0.03712/share. We have not observed an ask, bid or fill. Applying the full 4 bp to both a buy and a sell would double the intended spread allowance.

Discuss

1. If measured slippage already compares a fill with the midpoint, may this half-spread be added again? Explain the overlap.
2. What extra timestamped data would be needed to replace the allowance with an observed quoted spread?

Sources: [1], [3], [4]. See the references page.

ISSUE 05

When does adding slippage count the spread twice?

Slippage compares an execution price with a specified reference. A buy above that reference or a sell below it is adverse. An overnight price move alone is not measured slippage.

Our default allowance is 1 bp commission + 2 bp half-spread + 1 bp additional slippage = 4 bp per direction. Additional means beyond the spread. Alternatively, a 3 bp all-in price shortfall plus 1 bp commission also totals 4 bp.

$$b = b_{\text{commission}} + b_{\text{half}} + b_{\text{additional}}, \quad b = 1 + 2 + 1 = 4 \text{ bp}$$

Before running: Which calculation charges the same spread component twice?

```
rates = pd.Series({"Separate allowances": 1 + 2 + 1,
                  "All-in price shortfall": 1 + 3,
                  "Spread counted twice": 1 + 3 + 2})
pd.DataFrame({"Calculation": rates.index, "Rate (bp)": rates.values,
              "One-share cost (USD)": (price * rates.values / 10000).round(5)})
```

Calculation	Rate (bp)	One-share cost (USD)
Separate allowances	4	0.07424
All-in price shortfall	4	0.07424
Spread counted twice	6	0.11135

The first two alternatives imply USD 0.07424 per share at the January 11 reference Close. The third implies USD 0.11135; its extra 2 bp comes from double counting, not from evidence of worse execution.

Real slippage may be favorable or adverse and depends on side, reference time and fill price. The nonnegative 1 bp here is a chosen sensitivity input. It is not estimated from AAPL daily data.

Discuss

1. If a fill-to-midpoint shortfall is 3 bp, which component in the first calculation must it replace?
2. Could a next-day price improvement make measured slippage favorable? Specify the buy/sell side and the price reference before answering.

Sources: [3], [4], [5]. See the references page.

ISSUE 06

Which date carries the change in position?

A position can change at today's Close even though today's ending return was earned by yesterday's holding. Charge the trade on its execution date.

s is the 0-or-1 signal. p is stock exposure during the interval ending today; a is exposure just after today's trade. Both exposures are 0 or 1. Under next-Close execution, p uses the signal two observations earlier and a uses the signal one earlier. One-way turnover τ is the absolute change in stock exposure.

$$p_t = S_{t-2}, \quad a_t = S_{t-1}, \quad \tau_t = |a_t - p_t|$$

$$\tau_{\text{Jan 11}} = |1 - 0| = 1$$

Before running: Will the January 11 entry be charged on January 11 or January 12?

```
position = signal.shift(2).fillna(0)
after_close = signal.shift(1).fillna(0)
after_close.iloc[-1] = position.iloc[-1]
turnover = (after_close - position).abs()
ledger = pd.DataFrame({"Held": position, "After trade": after_close,
                      "Turnover": turnover})
shown = ledger.loc["2024-01-10":"2024-01-12"].copy()
shown.insert(0, "Date", shown.index)
shown
```

Date	Held	After trade	Turnover
2024-01-10	0.0	0.0	0.0
2024-01-11	0.0	1.0	1.0
2024-01-12	1.0	1.0	0.0

January 11 has held exposure 0, after-trade exposure 1 and turnover 1. January 12 has held exposure 1 and no new trade. Assigning costs to `position.diff()` would charge this entry a day late.

For a binary stock/cash account, exposure does not drift within either state. Partial multi-asset positions do drift; Week 8 will distinguish pretrade holdings from earlier targets. No final-Close trade is added here.

Discuss

1. Which date earns the first return from the January 11 entry? Point to the holding rather than the signal.
2. If a program subtracts a fee on every day Held equals 1, what is it charging incorrectly?

Sources: [5], [6]. See the references page.

ISSUE 07

Which entry and exit trades must be charged?

An initial purchase and a later sale are separate transactions. A round trip pays for both directions; remaining invested does not create another trade each day.

The first AAPL round trip enters January 11 and exits January 17. Its one-way turnovers are 1 and 1. At a constant 4 bp rate, the rates sum to 8 bp. This sum is not necessarily the compounded percentage wealth loss, because the two charges use different wealth bases.

$$\tau_{\text{entry}} + \tau_{\text{exit}} = 1 + 1 = 2, \quad 4(1 + 1) = 8 \text{ bp}$$

Before running: Does the January 17 sale incur a charge even though the account ends in cash?

```
trades = ledger.loc[turnover > 0].copy()
trades["Side"] = np.where(trades["After trade"] > trades["Held"], "Buy", "Sell")
trades["Cost (bp)"] = 4 * trades["Turnover"]
shown = trades.iloc[:2][["Side", "Turnover", "Cost (bp)"]].copy()
shown.insert(0, "Date", shown.index)
shown
```

Date	Side	Turnover	Cost (bp)
2024-01-11	Buy	1.0	4.0
2024-01-17	Sell	1.0	4.0

The full period contains 30 position-changing trades: 15 buys and 15 sells. The displayed first two transactions form one round trip. Their sum of cost rates is 8 bp, not one 4 bp charge.

Starting already invested would omit a purchase that starting in cash includes. A forced final sale would add a trade if stock remains held. State these boundary choices before comparing strategies.

Discuss

1. A buy-and-hold account starts in cash and is valued without a final sale. How many direction charges should it have?
2. Why can the exit fee in dollars differ from the entry fee even when both use 4 bp?

Sources: [2], [5]. See the references page.

ISSUE 08

How does a dated trade change net return?

Net return deducts the stated execution cost from the gross holding return on the same accounting date. The entry fee can reduce wealth before the first stock return is earned.

r is the decimal Close-to-Close stock return. The gross portfolio return is held exposure p times r . With $c = 4/10,000$, the cost deduction is cr of opening wealth. This linear approximation does not solve exact fee-funded holdings.

$$r_{p,t}^{\text{gross}} = p_t r_t, \quad r_{p,t}^{\text{net}} = p_t r_t - c \tau_t$$

$$r_{p,\text{Jan } 11}^{\text{net}} = 0 - 0.0004 \times 1 = -0.0004 = -0.04\%$$

Before running: Why should January 11 have negative net return despite zero stock exposure during its ending interval?

```
stock_return = close.pct_change(fill_method=None)
assert stock_return.iloc[1:].notna().all()
gross = position * stock_return.fillna(0)
cost_rate = 4 / 10000
net = gross - cost_rate * turnover
ledger["Gross"] = gross
ledger["Net"] = net
shown = pd.DataFrame({"Gross (%)": 100 * gross,
                      "Cost (bp)": 4 * turnover, "Net (%)": 100 * net})
shown = shown.loc[["2024-01-11", "2024-01-12", "2024-01-17"]].round(4)
shown.insert(0, "Date", shown.index)
shown
```

Date	Gross (%)	Cost (bp)	Net (%)
2024-01-11	-0.0	4.0	-0.04
2024-01-12	0.1778	0.0	0.1778
2024-01-17	-0.5174	4.0	-0.5574

January 11 is -0.04% net: the cash account pays for its closing purchase. January 17 earns -0.5174% gross before the sale, then pays 4 bp. The initial January 2 row is only an account boundary; it is not an observed zero stock return.

Discuss

1. A sale is moved from January 17 to January 18. Which holding and cost rows must be recalculated?
2. Why must a missing stock return while invested stop the calculation instead of being replaced by zero?

Sources: [1], [5], [6]. See the references page.

ISSUE 09

Why can the wealth gap differ from paid fees?

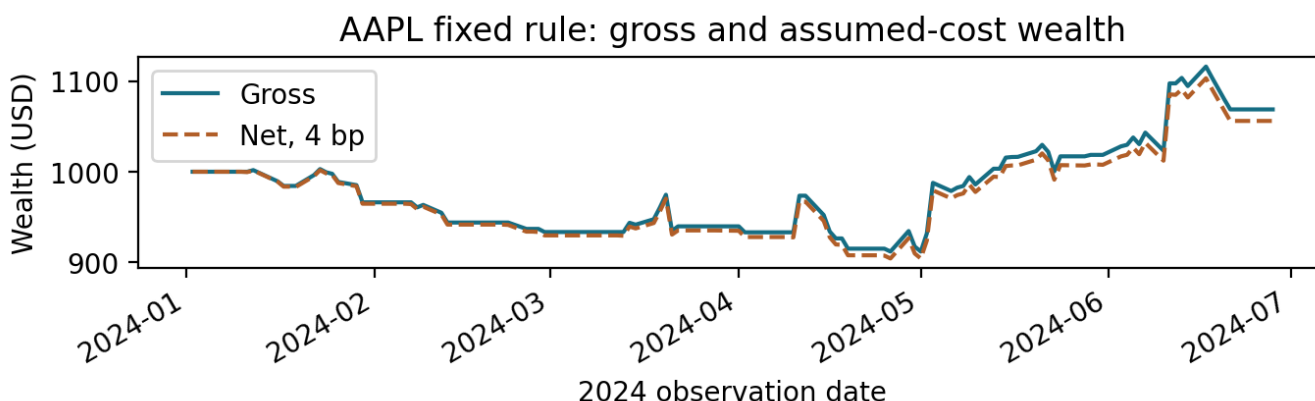
A fee reduces wealth available for later returns. The final gross-minus-net wealth gap therefore need not equal the sum of dollar fees paid.

W is wealth in USD and r is the corresponding gross or net portfolio return. Starting from USD 1,000, multiply the previous balance by each return factor. For the January 11 entry, net wealth becomes USD 999.60.

$$W_t = W_{t-1}(1 + r_{p,t}), \quad 1000(1 - 0.0004) = 999.60 \text{ USD}$$

Before running: Must the 4 bp wealth line stay at or below the gross line when all return factors remain positive?

```
wealth = pd.DataFrame({"Gross": 1000 * (1 + gross).cumprod(),
                      "Net, 4 bp": 1000 * (1 + net).cumprod()})
opening = wealth["Net, 4 bp"].shift(1).fillna(1000)
fees = opening * cost_rate * turnover
ax = wealth.set_axis(pd.to_datetime(wealth.index)).plot(
    figsize=(7, 2.3), style=["-", "--"], color=["#156d83", "#b15e28"])
ax.set(title="AAPL fixed rule: gross and assumed-cost wealth",
       xlabel="2024 observation date", ylabel="Wealth (USD)")
plt.tight_layout()
plt.gcf()
```



June 28: gross USD 1068.80, net USD 1056.07. The endpoint gap is USD 12.72; summed modeled fees are USD 11.60. Later returns act on the earlier deductions, so these amounts need not match.

Discuss

1. Why is subtracting all fees only at the endpoint not the same calculation? Identify what happens to the intervening investment base.
2. Could net exceed gross with nonnegative costs and positive gross/net factors under otherwise identical rules? Explain the product comparison.

Sources: [2], [5]. See the references page.

ISSUE 10

Can the entire cash balance buy stock before paying a fee?

Buying USD 1,000 of stock with USD 1,000 cash leaves no money for a fee. An exactly funded proportional-cost purchase must reserve cash for that fee.

Let q be shares, C the assumed execution price in USD/share, c the proportional fee rate and W available cash in USD. At the January 11 Close, the purchase and fee together must fit the budget. Fractional shares are allowed.

$$qC + cqC = W \implies q = \frac{W}{C(1+c)}$$

$$q \approx \frac{1000}{185.59000(1 + 0.0004)} \approx 5.386067 \text{ shares}$$

Before running: Which purchase needs more than the available USD 1,000?

```
q_without_fee = 1000 / price
q_with_fee = 1000 / (price * (1 + cost_rate))
q = np.array([q_without_fee, q_with_fee])
pd.DataFrame({"Calculation": ["No fee reserved", "Fee reserved"],
             "Shares": q.round(6),
             "Cash needed (USD)": (q * price * (1 + cost_rate)).round(4)})
```

Calculation	Shares	Cash needed (USD)
No fee reserved	5.388221	1000.4
Fee reserved	5.386067	1000.0

Ignoring the fee requires USD 1,000.40. Reserving it buys 5.386067 shares for total cash USD 1,000. The exact stock value is USD 999.600160; the main linear model records USD 999.600000.

The small difference is real: $1/(1+c)$ is not exactly $1-c$. The main model remains an explicit approximation so that we can isolate cost assumptions; it must not be described as an exact order-and-cash simulation.

Discuss

1. If c rises, does q increase or decrease under the fixed cash budget? Use the denominator.
2. When adapting this lesson into an executable trading simulator, what cash and holdings checks are needed beyond gross return minus cost?

Sources: [5]. See the references page.

ISSUE 11

What changes when execution is one Close later?

An additional execution delay changes which price intervals are held and when trades pay costs. It does not automatically reduce a historical return.

Keep signals and the 4 bp cost rate unchanged. Add one stored observation of delay. The held exposure becomes s from three observations earlier; after-trade exposure uses two earlier. Recalculate both positions and turnover.

$$p_t^{\text{delay}} = s_{t-3}, \quad a_t^{\text{delay}} = s_{t-2}$$

For the January 10 signal, the base entry is January 11 and the delayed entry is January 12. Their first returns end January 12 and January 16 respectively; weekends and the holiday are not extra observations.

$$a_{\text{Jan 12}}^{\text{delay}} = s_{\text{Jan 10}} = 1, \quad p_{\text{Jan 16}}^{\text{delay}} = 1$$

Before running: Must the later execution produce a smaller ending balance on this historical path?

```

delayed_position = signal.shift(3).fillna(0)
delayed_after = signal.shift(2).fillna(0)
delayed_after.iloc[-1] = delayed_position.iloc[-1]
delayed_turnover = (delayed_after - delayed_position).abs()
delayed_net = delayed_position * stock_return.fillna(0) - cost_rate * delayed_turnover
delayed_wealth = 1000 * (1 + delayed_net).cumprod()
pd.DataFrame({"Execution": ["Next Close", "One Close later"],
              "First entry": [turnover[turnover > 0].index[0],
                              delayed_turnover[delayed_turnover > 0].index[0]],
              "Final net (USD)": [round(wealth["Net, 4 bp"].iloc[-1], 2),
                                  round(delayed_wealth.iloc[-1], 2)]})

```

Execution	First entry	Final net (USD)
Next Close	2024-01-11	1056.07
One Close later	2024-01-12	1080.85

The delayed account ends at USD 1080.85, versus USD 1056.07 for the base execution. This is a different execution scenario, not measured slippage or evidence that delaying orders is generally beneficial.

Discuss

1. Why is shifting returns while retaining the old cost dates not a valid delayed implementation?
2. Could the two schedules have equal total turnover but different net wealth? Identify the changed dates that still matter.

Sources: [4], [6]. See the references page.

ISSUE 12

Does trading less often necessarily save more money?

Checking a signal every fifth observation may reduce trading, but it also changes exposure. A larger net balance cannot automatically be attributed only to saved fees.

Compute the same daily signal. Allow a new decision only at indices 0, 5, 10 and so on, carrying it forward between those dates. This is a five-observation schedule, not a calendar-week rule. Execution remains at the next Close.

Before running: Will the less frequent schedule have the same gross wealth as the daily schedule?

```
scheduled_signal = signal.iloc[::5].reindex(signal.index).ffill()
scheduled_position = scheduled_signal.shift(2).fillna(0)
scheduled_after = scheduled_signal.shift(1).fillna(0)
scheduled_after.iloc[-1] = scheduled_position.iloc[-1]
scheduled_turnover = (scheduled_after - scheduled_position).abs()
scheduled_gross = scheduled_position * stock_return.fillna(0)
scheduled_net = scheduled_gross - cost_rate * scheduled_turnover
pd.DataFrame({"Decision schedule": ["Daily", "Every fifth row"],
             "Turnover": [turnover.sum(), scheduled_turnover.sum()],
             "Gross (USD)": [wealth.Gross.iloc[-1], 1000 * (1 + scheduled_gross).prod()],
             "Net (USD)": [wealth["Net, 4 bp"].iloc[-1], 1000 * (1 + scheduled_net).prod()]
            }).round(2)
```

Decision schedule	Turnover	Gross (USD)	Net (USD)
Daily	30.0	1068.8	1056.07
Every fifth row	18.0	1017.4	1010.15

Total turnover changes from 30 to 18; gross wealth changes from USD 1068.80 to USD 1017.40. Both market exposure and costs change. The result does not isolate fees alone.

Discuss

1. Which comparison would isolate only the cost rate while keeping exposure unchanged?
2. Does every fifth observation always produce exactly one trade? Use the distinction between checking a signal and changing a position.

Sources: [5], [6]. See the references page.

ISSUE 13

How sensitive is wealth to the assumed cost rate?

To isolate the cost assumption, keep the data, signals, positions, execution dates and terminal treatment fixed. Change only the cost rate.

For each assumed rate b in bp, recalculate every net return and compound the resulting wealth. With nonnegative turnover and positive return factors, a higher rate cannot increase wealth under this fixed strategy.

$$W_T(b) = 1000 \prod_t \left(1 + r_{p,t}^{\text{gross}} - \tau_{t \frac{b}{10000}} \right)$$

$$W_{\text{Jan } 11}(10) = 1000(1 + 0 - 1 \times 0.001) = 999.00 \text{ USD}$$

Before running: Can the ending balance increase as the rate rises if everything else stays fixed?

```
sensitivity = []
for bp in [0, 4, 10, 20, 40]:
    factors = 1 + gross - turnover * bp / 10000
    assert (factors > 0).all()
    sensitivity.append([bp, 1000 * factors.prod()])
sensitivity = pd.DataFrame(sensitivity, columns=["Rate (bp)", "Final net (USD)"])
sensitivity.round(2)
```

Rate (bp)	Final net (USD)
0	1068.8
4	1056.07
10	1037.26
20	1006.63
40	947.96

At 0 bp the ending balance is USD 1068.80; at 40 bp it is USD 947.96. This checks sensitivity to an assumed fee. It does not identify a realistic historical rate from daily prices.

Discuss

1. If a 40 bp run ends above its 4 bp counterpart, which fixed-position or accounting conditions should be checked first?
2. Would choosing a different signal threshold for every cost rate still isolate the cost assumption? State what else changes.

Sources: [2], [5]. See the references page.

ISSUE 14

What cost rate removes the historical gain?

A historical break-even cost is the assumed rate that makes final net wealth equal the chosen benchmark. Here the benchmark is USD 1,000 cash earning zero interest.

The rate b is in bp and $W(b)$ is the compounded balance from Issue 13. We solve $W(b) = 1,000$ inside a bracket where the result changes sides. This is not the simple gross gain divided by total turnover: costs compound on different dates.

$$W(0) \approx 1068.80 > 1000, \quad W(40) \approx 947.96 < 1000$$

Before running: Does the bracket imply a crossing within 0-40 bp under this continuous fixed-path model?

```
lo, hi = 0.0, 40.0
assert 1000 * (1 + gross - turnover * lo / 10000).prod() > 1000
assert 1000 * (1 + gross - turnover * hi / 10000).prod() < 1000
for _ in range(50):
    mid = (lo + hi) / 2
    value = 1000 * (1 + gross - turnover * mid / 10000).prod()
    if value > 1000:
        lo = mid
    else:
        hi = mid
break_even = (lo + hi) / 2
pd.DataFrame({"Break-even (bp)": [round(break_even, 4)],
             "Final net (USD)": [round(value, 4)]})
```

Break-even (bp)	Final net (USD)
22.2013	1000.0

The computed rate is 22.2013 bp per direction. It applies only to this AAPL period, rule, cash benchmark and linear cost approximation. It does not estimate actual costs or establish that a future gain can absorb this rate.

Discuss

1. If both bracket endpoints remained above USD 1,000, could this code honestly report a crossing inside the bracket?
2. Why would comparing against buy-and-hold require solving a different equation, including the benchmark's own cost?

Sources: [2], [5]. See the references page.

ISSUE 15

How should cash and buy-and-hold be charged?

Cash tests the choice not to hold stock. Buy-and-hold tests continuous exposure. Both need the same dates and an explicit initial purchase and final-sale convention.

Cash starts at USD 1,000 and earns zero. Buy-and-hold purchases at the first Close, pays the 4 bp opening charge, then makes no further trade. No forced final sale is assumed. Its net entry value is USD 999.60 under the same linear approximation.

$$W_T^{BH} = 1000(1 - c) \frac{C_T}{C_0}$$

$$W_T^{BH} \approx 1000(1 - 0.0004) \frac{210.62000}{185.64000} \approx 1134.11 \text{ USD}$$

Before running: Should buy-and-hold pay 4 bp on every day, once at entry, or once on each entry and assumed final sale?

```
buy_hold_wealth = 1000 * (1 - cost_rate) * close / close.iloc[0]
benchmark_summary = pd.DataFrame({"Strategy": ["Cash", "Buy-and-hold", "Fixed rule"],
    "Turnover": [0, 1, turnover.sum()],
    "Final net (USD)": [1000, buy_hold_wealth.iloc[-1], wealth["Net, 4 bp"].iloc[-1]]})
benchmark_summary.round(2)
```

Strategy	Turnover	Final net (USD)
Cash	0.0	1000.0
Buy-and-hold	1.0	1134.11
Fixed rule	30.0	1056.07

Buy-and-hold ends at USD 1134.11 after one charge. The fixed rule has 30 units of turnover. Those different exposures and trades are part of the comparison, not quantities to force equal.

The January 2 buy-and-hold fee is an initial boundary charge, not a close-to-close market return. Keep it in total wealth, but distinguish it from the later daily-return observations when computing Week 7 statistics.

Discuss

1. What changes if the report requires all strategies to finish in cash? Specify the additional trade, rather than silently subtracting another fee.
2. Does the larger endpoint alone establish lower risk or more reliable future returns? Name the evidence not shown here.

Sources: [2], [5]. See the references page.

ISSUE 16

What must be checked before interpreting a net backtest?

Before interpreting a net result, check the dates and identities that produced it. A plausible final balance cannot reveal whether an entry fee was charged a day late.

Use the actual entry, exit and final valuation rows. Verify that today's holding equals the preceding after-trade holding, that net return deducts only the stated cost, and that no negative wealth factor is hidden.

Before running: Which displayed date must have an entry fee but no stock return earned yet?

```
assert position.iloc[1:].equals(after_close.shift(1).iloc[1:])
assert (turnover >= 0).all() and ((1 + net) > 0).all()
assert np.allclose(net, gross - cost_rate * turnover)
audit = pd.DataFrame({"Fee (USD)": fees,
    "Gross wealth (USD)": wealth.Gross, "Net wealth (USD)": wealth["Net, 4 bp"]})
audit = audit.loc[["2024-01-11", "2024-01-17", "2024-06-28"]].round(4)
audit.insert(0, "Date", audit.index)
audit
```

Date	Fee (USD)	Gross wealth (USD)	Net wealth (USD)
2024-01-11	0.4	1000.0	999.6
2024-01-17	0.3956	984.3203	983.5309
2024-06-28	0.0	1068.7988	1056.0744

January 11 has a USD 0.4000 fee and net wealth USD 999.6000. January 17 has its own exit charge. These checks support this accounting calculation; they do not verify that real orders filled or that 4 bp describes their actual costs.

Week 7 keeps this same 4 bp rule and evaluates return, risk and uncertainty. Do not choose a weighting method or rewrite the strategy while comparing its performance measures.

Discuss

1. A new version has the same final gross balance but shifts all cost dates by one observation. Which assertion or dated row would expose it?
2. A report says this strategy is profitable in practice because the ending balance exceeds USD 1,000. Rewrite the claim to state the tested dates, cost assumption and missing execution evidence.

Sources: [1], [4], [5], [6]. See the references page.

REFERENCES AND REPRODUCTION

Sources and limits

[1] [yfinance](#): historical price fields

[2] [Investor.gov](#): understanding fees

[3] [Investor.gov](#): bid and ask

[4] [Investor.gov](#): executing an order

[5] [Lobo, Fazel and Boyd \(2007\)](#): linear and fixed transaction costs

[6] [pandas](#): shifting observations

All commissions, spreads and slippage allowances are explicit hypothetical inputs. No broker, historical quote or fill is being estimated. The linear account approximation excludes market impact, taxes, borrowing, financing, partial fills and dividend cash flows. Its conclusions are conditional on the stated dates and rules.

Executed with Python 3.12.9, NumPy 2.2.6, pandas 2.2.3, matplotlib 3.10.8, yfinance 1.2.0. Run displayed code in order in a fresh Python session. Data requests need internet access; historical provider responses may change.

Selected cases illustrate the stated calculations, not investment recommendations or future-performance guarantees. Full snapshots and instructor checks are not included in the public PDF.